

ENSO Intensity and California Statewide Precipitation: Water-Year Analysis

DATASCI 203 — Group 2

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1 Data

1.1 The ENSO series

The ONI file is pre-built: the December-February mean, NOAA’s phase and strength classifications, 0/1 phase indicators, a one-year lag, and a linear time trend. No derivation is needed here, which removes a whole class of possible error - but it also means the water-year alignment is inherited rather than verified, so the checks below matter.

Table 1: ENSO series validation.

Check	Value
Water years	76
Range	WY1950-WY2025
Mean ONI	0.01
SD ONI	1.04
Phase counts (La Nina / Neutral / El Nino)	29 / 20 / 27
Lag available for	75 years

1.2 The precipitation series

The preferred outcome is NOAA’s statewide water-year total. If that file is not present, the chunk below falls back to a station-weighted composite of the regional wet-season data, and flags the substitution loudly. **The fallback is not the same construct** — it covers November through April, so it runs below the statewide annual figure, and the report must say so wherever levels are quoted.

1.3 Merge

All six models are estimated on these 73 water years. Restricting to rows with a defined lag costs the first sample year but makes AIC and the nested F-test legitimate comparisons rather than apples-to-oranges.

2 Descriptive analysis

Table 2: Descriptive statistics for the analysis variables.

Variable	N	Mean	SD	Min	Median	Max
Precipitation (in)	73	22.12	6.18	7.93	21.76	42.46
ONI (Dec-Feb, C)	73	0.04	1.03	-1.80	-0.10	2.60
Previous winter ONI (C)	73	0.05	1.03	-1.80	-0.10	2.60
Reversed SOI (Nov-Mar)	73	-0.30	1.52	-3.88	-0.22	4.76
Previous winter reversed SOI	73	-0.27	1.52	-3.88	-0.20	4.76

Table 3: Precipitation by ENSO phase.

ENSO phase	Water years	Mean precip (in)	SD (in)	Driest	Wettest
Neutral	20	21.6	5.4	7.9	32.0
El Nino	26	24.8	7.5	12.1	42.5
La Nina	27	20.0	4.3	11.8	32.5

Table 4: Precipitation by ENSO strength bin. The very strong bin holds only a handful of years and cannot support inference on its own.

Strength	Water years	Mean precip (in)	SD (in)
neutral	20	21.6	5.4
weak	30	21.4	5.3
moderate	9	21.8	6.7
strong	11	22.0	6.1
very_strong	3	34.9	8.5

Welch two-sample t-test, El Nino years vs all others: $t = 2.52$, $df = 36.8$, $p = 0.016$

2.1 Figures

3 Models

Model	Question
M1	Is there a linear ENSO-precipitation association?
M2	What does collapsing intensity into phase categories cost?
M3	Is the association confounded by last winter's ENSO or by trend?
M4	Is the relationship proportional rather than additive?
M5	Does the relationship steepen at extreme ONI values?

Model	Question
M6	Does nesting the continuous rSOI add anything beyond the baseline?

Table 6: ENSO intensity and California water-year precipitation.

	<i>Dependent variable:</i>					
	M1	precip_in M2	M3	log(precip_in) M4	precip_in M5	M6
Winter ONI	2.714*** (0.661)		2.786*** (0.684)	0.113*** (0.026)	2.416*** (0.603)	2.582* (1.514)
El Nino						0.693 (1.046)
La Nina		3.196* (1.898)				-1.965 (2.603)
Previous winter ONI		-1.560 (1.464)				
Time trend			0.727 (0.579)			
Winter ONI squared			-0.007 (0.029)			
Reversed SOI					0.668 (0.552)	
Constant	22.019*** (0.640)	21.562*** (1.203)	22.259*** (1.141)	3.053*** (0.031)	21.323*** (0.881)	22.934*** (1.182)
Observations	73	73	73	73	73	73
R ²	0.207	0.112	0.222	0.166	0.226	0.223
Adjusted R ²	0.196	0.087	0.188	0.154	0.204	0.189

Note: *p<0.1; **p<0.05; ***p<0.01
HC1 heteroskedasticity-robust standard errors.

3.1 Practical significance

Table 7: Effect sizes implied by Model 1. The final column is the one that matters for planning: it compares the ENSO signal to the year-to-year noise it must be detected against.

Scenario	Change vs mean (in)	As pct of mean	As pct of SD
One SD increase in winter ONI	2.81	12.70	45.48

Strong El Nino (ONI = +1.5)	4.07	18.40	65.91
Very strong El Nino (ONI = +2.3)	6.24	28.22	101.07
Strong La Nina (ONI = -1.5)	-4.07	-18.40	-65.91

3.2 Model comparison

Table 8: M4's outcome is on a different scale, so its R-squared and AIC are NOT comparable to the others. Compare it only to itself.

Model	R-squared	Adj R-squared	AIC
M1 linear	0.2068	0.1956	461.0713
M2 phase dummies	0.1122	0.0869	471.2954
M3 lag + trend	0.2222	0.1884	463.6386
M4 log outcome	0.1658	0.1541	16.7047
M5 quadratic	0.2259	0.2038	461.2936
M6 ONI + SOI + El Nino	0.2230	0.1893	463.5612

```
## Analysis of Variance Table
##
## Model 1: precip_in ~ oni_djf
## Model 2: precip_in ~ oni_djf + oni_djf_lag1 + time_trend
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      71 2178.6
## 2      69 2136.3  2    42.338 0.6838 0.5081

## Analysis of Variance Table
##
## Model 1: precip_in ~ oni_djf
## Model 2: precip_in ~ oni_djf + I(oni_djf^2)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      71 2178.6
## 2      70 2126.2  1    52.41 1.7255 0.1933
```

4 Regression assumptions

Table 9: Classical linear model assumption checks.

Assumption	Test	Value
IID / no serial correlation	Breusch-Godfrey(2)	LM = 1.72, p = 0.422
IID / no serial correlation	Lag-1 residual autocorrelation	0.071
Homoskedasticity (levels)	Breusch-Pagan	p = 0.028
Homoskedasticity (log)	Breusch-Pagan	p = 0.620
Normal errors (levels)	Shapiro-Wilk	p = 0.950
Normal errors (log)	Shapiro-Wilk	p = 0.022

Skewness (levels)	-	0.59
Skewness (log)	-	-0.43

Table 10: Variance inflation factors, M3.

Term	VIF
oni_djf	1.01
oni_djf_lag1	1.01
time_trend	1.00

5 Reproducibility check

Table 11: Values computed independently in Python from the same two source files. The descriptive rows also match the published report exactly, which confirms the pipeline reproduces the original analysis. If the knit disagrees with any row here, something drifted.

Quantity	Expected
Descriptives n = 76 (WY1950-2025)	mean 22.18, SD 6.75, min 10.86, median 20.80, max 40.41
Correlation with ONI	0.257
Skewness levels -> log	0.65 -> 0.11
Phase groups (La Nina/Neutral/El Nino)	29 / 20 / 27
Welch t-test, El Nino vs rest	t = 1.77, p = 0.084
Model sample	n = 75 (WY1951-2025)
M1 slope	1.645 (HC1 SE 0.807, p = 0.042)
M1 intercept / R-squared	22.177 / 0.0620
M2 El Nino / La Nina	3.251 (p = 0.120) / 0.348 (p = 0.840)
M2 R-squared	0.0477 (below M1)
M3 ONI / lag / trend	1.765 (p = 0.028) / 1.379 (p = 0.047) / -0.027 (p = 0.425)
M4 semi-elasticity	0.062 (p = 0.061)
M5 quadratic term	1.011 (p = 0.122)
Nested F, M1 vs M3	2.03, p = 0.139
Nested F, M1 vs M5	2.81, p = 0.098
Breusch-Godfrey(2)	LM = 1.50, p = 0.471
Lag-1 residual autocorrelation	-0.098
Breusch-Pagan (levels / log)	p = 0.010 / p = 0.170
Shapiro-Wilk (levels / log)	p = 0.009 / p = 0.043

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 24.04.4 LTS
##
```

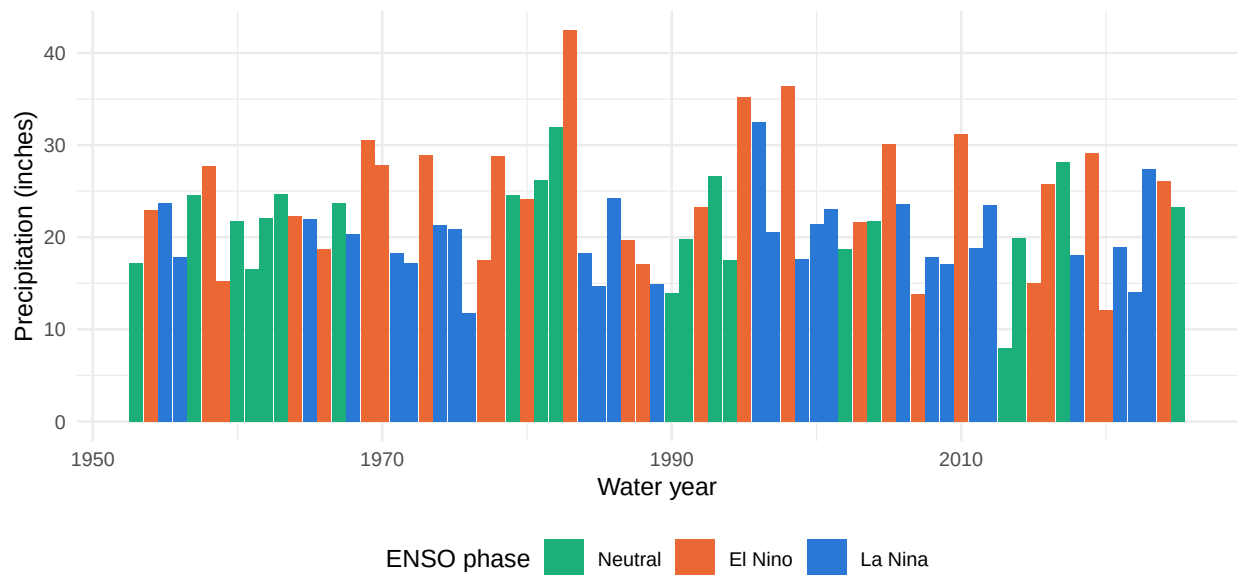


Figure 1: Statewide water-year precipitation, coloured by ENSO phase.

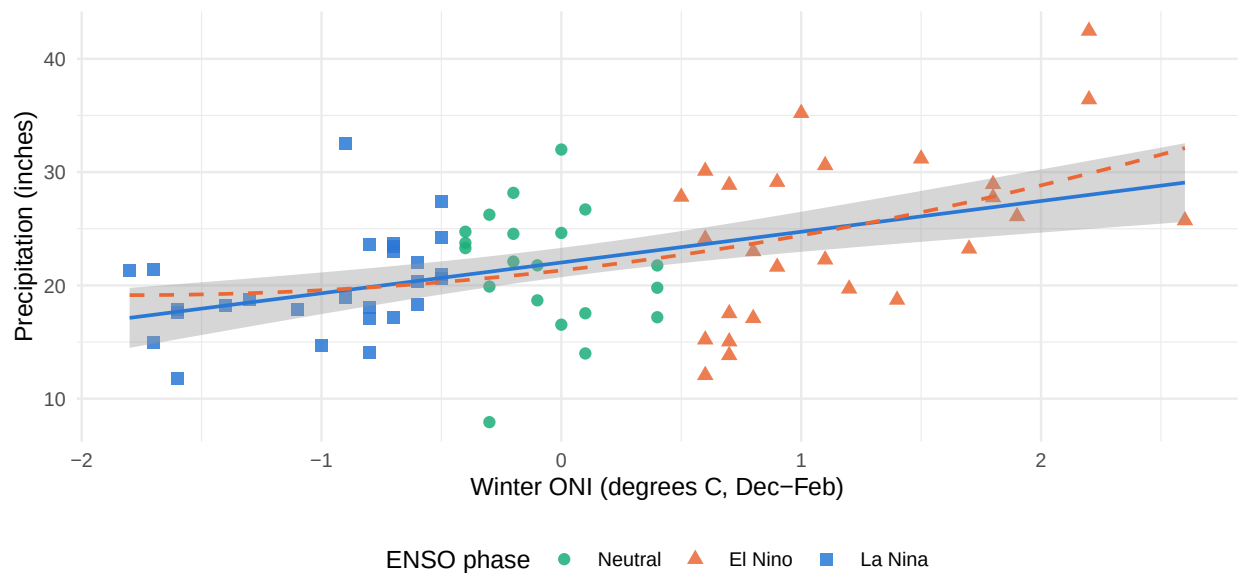


Figure 2: Precipitation against winter ONI, with linear and quadratic fits.

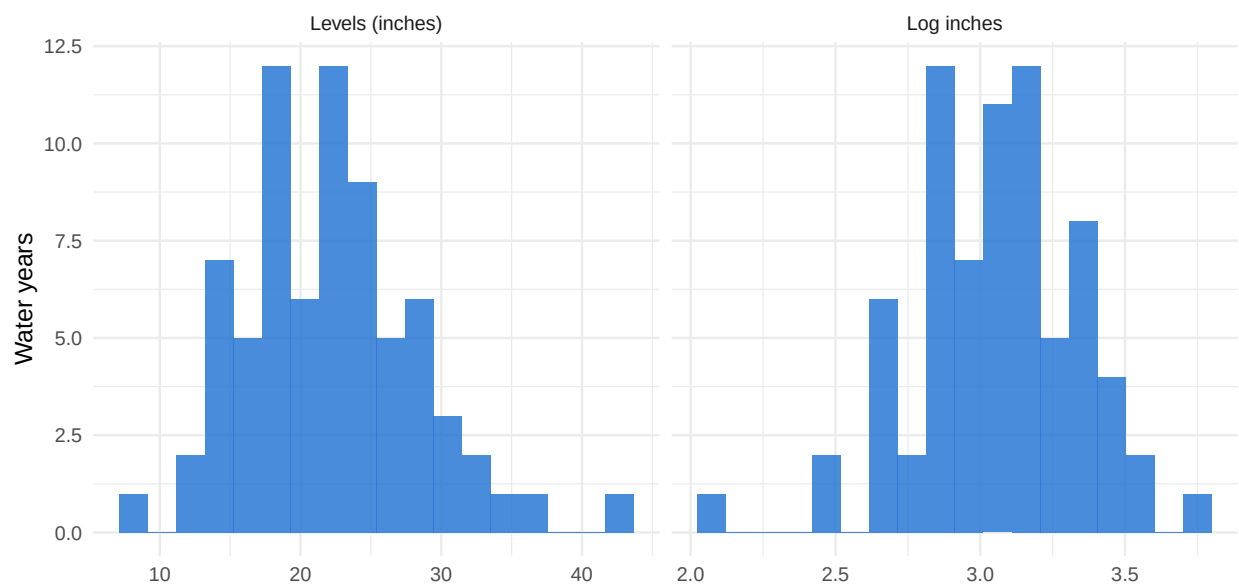


Figure 3: Precipitation is right-skewed; the log transform largely corrects it.

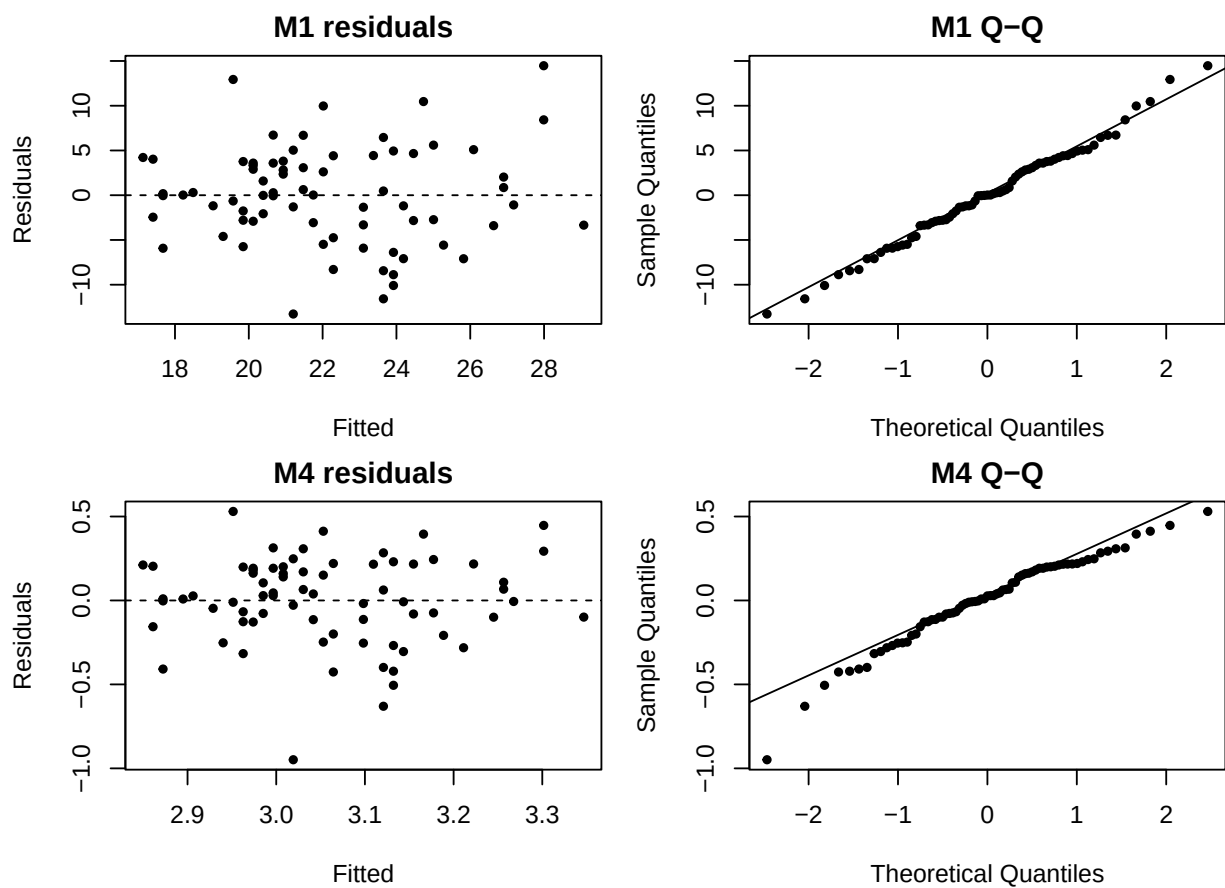


Figure 4: Diagnostic plots for Models 1 and 4.

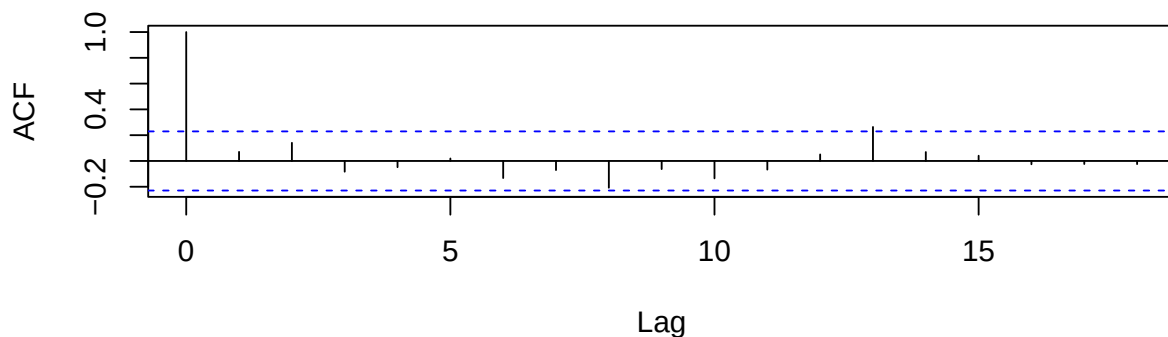


Figure 5: Residual autocorrelation, Model 1.

```
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] moments_0.14.1  kableExtra_1.4.0  knitr_1.45      car_3.1-2
## [5] carData_3.0-5   stargazer_5.2.3   lmtest_0.9-40   zoo_1.8-12
## [9] sandwich_3.1-0  lubridate_1.9.3   forcats_1.0.0   stringr_1.5.1
## [13] dplyr_1.1.4     purrr_1.0.2       readr_2.1.5     tidyr_1.3.1
## [17] tibble_3.2.1    ggplot2_3.4.4     tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] utf8_1.2.4      generics_0.1.3    xml2_1.3.6       stringi_1.8.3
## [5] lattice_0.22-5  hms_1.1.3         digest_0.6.34    magrittr_2.0.3
## [9] evaluate_0.23   grid_4.3.3        timechange_0.3.0 fastmap_1.1.1
## [13] Matrix_1.6-5    rprojroot_2.0.4   mgcv_1.9-1       fansi_1.0.5
## [17] viridisLite_0.4.2 scales_1.3.0      abind_1.4-5      cli_3.6.2
## [21] crayon_1.5.2    rlang_1.1.3       splines_4.3.3    bit64_4.0.5
## [25] munsell_0.5.0   withr_2.5.0       yaml_2.3.8       parallel_4.3.3
## [29] tools_4.3.3     tzdb_0.4.0        colorspace_2.1-0 vctrs_0.6.5
## [33] R6_2.5.1        lifecycle_1.0.4   bit_4.0.5        vroom_1.6.5
```



```
## [37] pkgconfig_2.0.3    pillar_1.9.0      gtable_0.3.4      glue_1.7.0
## [41] systemfonts_1.0.5  xfun_0.41         tidyselect_1.2.0  rstudioapi_0.15.0
## [45] farver_2.1.1       nlme_3.1-164      htmltools_0.5.7   labeling_0.4.3
## [49] rmarkdown_2.25     svglite_2.1.3     compiler_4.3.3
```